



RESEARCH ARTICLE SUMMARY

CANCER IMMUNOLOGY

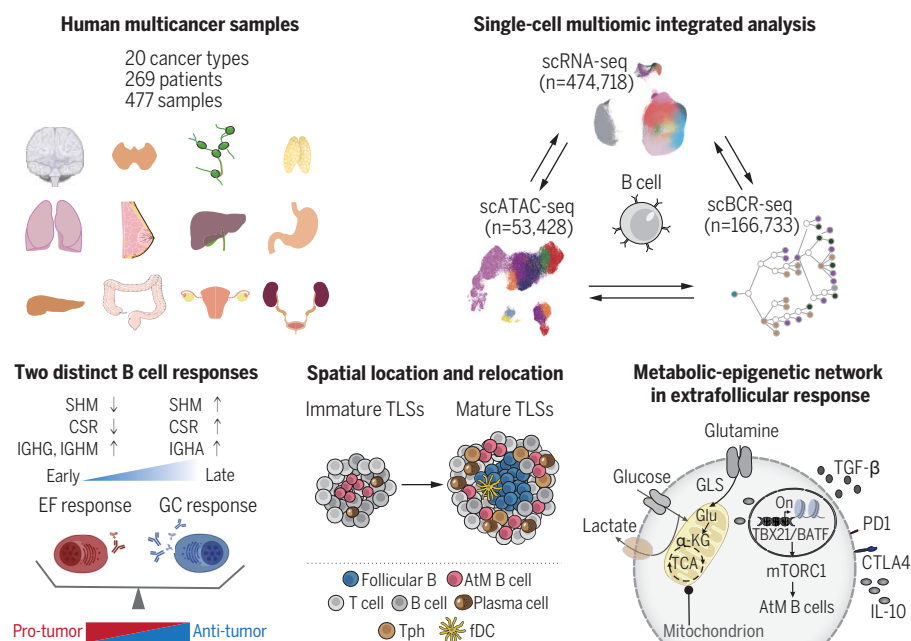
A blueprint for tumor-infiltrating B cells across human cancers

Jiaqiang Ma[†], Yingcheng Wu[†], Lifeng Ma[†], Xupeng Yang[†], Tiancheng Zhang[†], Guohe Song, Teng Li, Ke Gao, Xia Shen, Jian Lin, Yamin Chen, Xiaoshan Liu, Yuting Fu, Xixi Gu, Zechuan Chen, Shan Jiang, Dongning Rao, Jiaomeng Pan, Shu Zhang, Jian Zhou, Chen Huang, Si Shi, Jia Fan*, Guoji Guo*, Xiaoming Zhang*, Qiang Gao*

INTRODUCTION: Tumor-infiltrating B cells have emerged as important players in cancer immunity and served as predictors of response to immunotherapy. These B cells display multiple functions, primarily through their ability to differentiate into plasma cells to produce antibodies, but vary spatiotemporally across different cancer types. Dissecting the abundance and differentiation states of B cells across diverse cancer types holds promise for improving the immunotherapeutic response.

RATIONALE: To compile a comprehensive pan-cancer B cell landscape, we performed single-cell RNA sequencing (scRNA-seq) on paired tumors, lymph node metastases, adjacent normal tissues, and peripheral blood from

patients with various cancer types, as well as incorporating substantial published scRNA-seq datasets. After correction of the batch effect, this atlas consists of scRNA-seq data from 269 patients across 20 cancer types. We assembled B cell receptor (BCR) sequencing of individual B cells with gene-expression profiles to characterize the dynamic transition between B cells and antibody-secreting cells (ASCs). We integrated the single-cell chromatin accessibility landscape of B cells from different cancers to dissect the epigenomic regulation networks that function in fine-tuning B cell development. We spatially localized B cells in mature versus immature tertiary lymphoid structures (TLSs) and investigated the potential regulators that direct B cells into specific responses.



Systematic analysis of a human pan-cancer B cell atlas. We analyzed 474,718 B cells from 269 patients across 20 cancer types using single-cell sequencing data. By combining gene expression profiles, BCR sequences, and chromatin accessibility, we investigated the diversity and plasticity of tumor-infiltrating B cells and performed a multilevel comparison of EF- and GC-responsive plasma cells among cancer types. We visualized their dynamic spatial locations along the maturation of TLSs and identified potential metabolic-epigenetic mechanisms in regulating B cell differentiation.

RESULTS: We revealed substantial heterogeneity within B and plasma cells, identifying B cell subsets and 10 plasma cell subsets. We computationally derived and validated two independent developmental pathways to ASCs through canonical germinal center (GC) and alternative extrafollicular (EF) pathways and demonstrated an apparent cancer-type preference. Colon adenocarcinoma and liver hepatocellular carcinoma were the two representative types of cancer enriched for GC and EF pathways, respectively. We affirmed that EF-dominant cancers correlate with dysregulated immune responses and worse clinical outcomes. We then identified the dynamic metabolic-epigenetic-signaling networks engaged in fine-tuning tumor-infiltrating B cell differentiation and managing the balance between the EF and GC pathways. Atypical memory (AtM) B cells, the primary progenitors of EF-derived ASCs, exhibit an exhausted and bystander phenotype and develop independently of the GC pathway. We found that the AtM B cells reside in the center of immature TLSs and spatially relocate to the periphery during TLS maturation. Last, we mechanistically linked these findings to specific transcription factors and epigenomic regulations. We demonstrated that the glutamine-derived metabolite α -ketoglutarate (α -KG) could increase the expression of AtM B cell-associated transcription factors *T-bet* and *BATF* and promote their differentiation, accompanied by the activation of mammalian target of rapamycin complex 1 (mTORC1) signaling. Consequently, AtM B cells acquire an immunoregulatory function that dampens antitumor T cell responses and fosters an immunosuppressive microenvironment.

CONCLUSION: We compiled the blueprint of B cell heterogeneity and two dynamic differentiation pathways in human cancers, providing a fundamental reference of ASC differentiation trajectory for future studies. The systematic comparison between EF and GC pathways reveals the similarities and differences of B cell states across different cancer types, highlighting the unfavorable clinical outcome linked to the immunosuppressive microenvironment of EF pathway-associated AtM B cells. Metabolic-epigenetic networks are remarkably flexible and can reconfigure B cell fates in a way that will facilitate the development of B cell-targeted immunotherapies. ■

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Cite this article as J. Ma et al., *Science* **384**, eadj4857 (2024). DOI: 10.1126/science.adj4857

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B lymphocytes are essential mediators of humoral immunity and play multiple roles in human cancer. To decode the functions of tumor-infiltrating B cells, we generated a B cell blueprint encompassing single-cell transcriptome, B cell–receptor repertoire, and chromatin accessibility data across 20 different cancer types (477 samples, 269 patients). B cells harbored extraordinary heterogeneity and comprised 15 subsets, which could be grouped into two independent developmental paths (extrafollicular versus germinal center). Tumor types grouped into the extrafollicular pathway were linked with worse clinical outcomes and resistance to immunotherapy. The dysfunctional extrafollicular program was associated with glutamine-derived metabolites through epigenetic-metabolic cross-talk, which promoted a T cell–driven immunosuppressive program. These data suggest an intratumor B cell balance between extrafollicular and germinal-center responses and suggest that humoral immunity could possibly be harnessed for B cell–targeting immunotherapy.

B cells are essential components of the adaptive immune system, playing central roles in the humoral immune response and antibody generation (1). Within the tumor microenvironment (TME), tumor-infiltrating B cells (TIBs) display considerable functional heterogeneity, broadly spanning naïve B cells, memory B cells (Bm), germinal center (GC) B cells, and antibody-secreting cells [ASCs, including plasmablasts (PBs) and plasma cells (PCs)], primarily on the basis of their immunophenotypes (2, 3). However, the current paradigm of B cell research is largely hypothesis-driven and may not unbiasedly capture the full spectrum of B cell states within the tumor milieu. With a specific focus on antitumor capabilities of producing antibodies against tumor-associated antigens, facilitating phagocytosis, and presenting antigens to CD8⁺ T cells, recent reports suggest that B cells may be used for next-generation immunotherapy (4–6). Spatially, these cells densely interact with T cells

within tertiary lymphoid structures (TLSs), which are associated with improved survival and hot tumor environment in several cancers (7–10). However, TIBs also exhibited tumor-promoting properties by releasing cytokines (11, 12), forming immune complexes (13, 14), and engaging immune checkpoints (15). These observations highlight the yin and yang roles of B cells and underscore the need for comprehensive data-driven analysis of B cells across human cancers.

Historically, ASCs were long believed to originate from the GC response. Under such a scenario, B cells encounter antigens, undergo somatic hypermutations, class-switch recombination (CSR), and clonal expansion, and eventually differentiate into long-lived PCs and generate high-affinity antibodies (16, 17). Alternatively, the recent discovery of extrafollicular (EF) differentiation highlighted short-lived ASCs, resulting in low-affinity and polyreactive antibodies (18–23). EF-associated B cells, also documented as age-associated B cells (ABCs), double-negative B cells (B_{DN}), or atypical memory (AtM) B cells in aging mice, autoimmune diseases, and chronic infection models, were characterized by CD21[−]CD11c⁺ and by expression of T-box transcription factor (*T-bet*) (19, 20, 24). We and other independent groups also reported signals such as Toll-like receptor 7/9 (TLR7/9), interferon- γ (IFN γ), interleukin-21 (IL-21), and CD40 in driving the development of EF-associated B cell states in autoimmune diseases (21, 22, 24). These data raise a critical question of what B cell fates are composed of and how they can be precisely regulated in cancer ecosystems. Systematically decoding the developmental hierarchy of B cells

in cancer may therefore delineate tumor-specific patterns as well as the clonal clades.

In this study, we compiled an atlas of TIBs across 20 different human cancers. We systematically characterized 15 B cell subsets and identified previously unrecognized AtM B cells and the presence of the EF pathway in the TME. Our data revealed two distinct ASC differentiation pathways—GC and EF paths—exhibiting cancer-type preferences. We characterized the distinctive phenotype, function, TLS localization, and clinical significance of AtM B cells in contrast to conventional GC B cells. In particular, tumor-infiltrating AtM B cells, the progenitors of EF-derived ASCs, exhibited an exhausted and bystander phenotype and developed independently of the GC. We linked these findings to specific transcription factors and epigenomic regulation and demonstrated the influence of the metabolic microenvironment, with glutamine playing a role in regulating AtM B cell differentiation and adopting an immunoregulatory function. Our study provides unprecedented large-scale B cell transcriptome data as a valuable resource for studying TIBs and potentially guiding effective B cell–directed cancer immunotherapeutic strategies.

Results

A single B cell transcriptome blueprint across human cancers

B cell infiltration varies highly across human cancer types (1). To understand the overall infiltration pattern of B cells, we assessed the abundance of B cell infiltration in The Cancer Genome Atlas (TCGA) Pan-Cancer datasets ($n = 8863$ tumor samples of 31 cancer types) on the basis of the consensus of six existing deconvolution algorithms (fig. S1A). We selected the cancer types with high and medium intratumor B cell scores for further sampling. CD19⁺ B cells were sorted from 153 samples of 66 patients across 15 human cancer types covering matched tumors, lymph node metastases (LN_Mets), adjacent normal tissues, and peripheral blood, followed by paired single-cell RNA sequencing (scRNA-seq) and single-cell B cell–receptor sequencing (scBCR-seq) (Fig. 1A and fig. S1B). We also integrated diverse published datasets containing TIBs, corrected the batch effects using the Harmony algorithm (25), and finally established a single-cell transcriptional atlas in 477 samples from 269 donors across 20 cancer types. After stringent quality control, we obtained a total of 474,718 single B cell transcriptomes, 69.24% of which were newly generated in this study (fig. S1C and table S1). For scBCR-seq, a total of 166,733 cells from 150,949 clonotypes spanning 61 donors of 15 cancer types were newly produced, carrying at least one pair of productive heavy and light chains, of which 12.76% were clonal ($\geq$ two B cells containing the identical BCR pair), corresponding to 21,268 expanded clonotypes.

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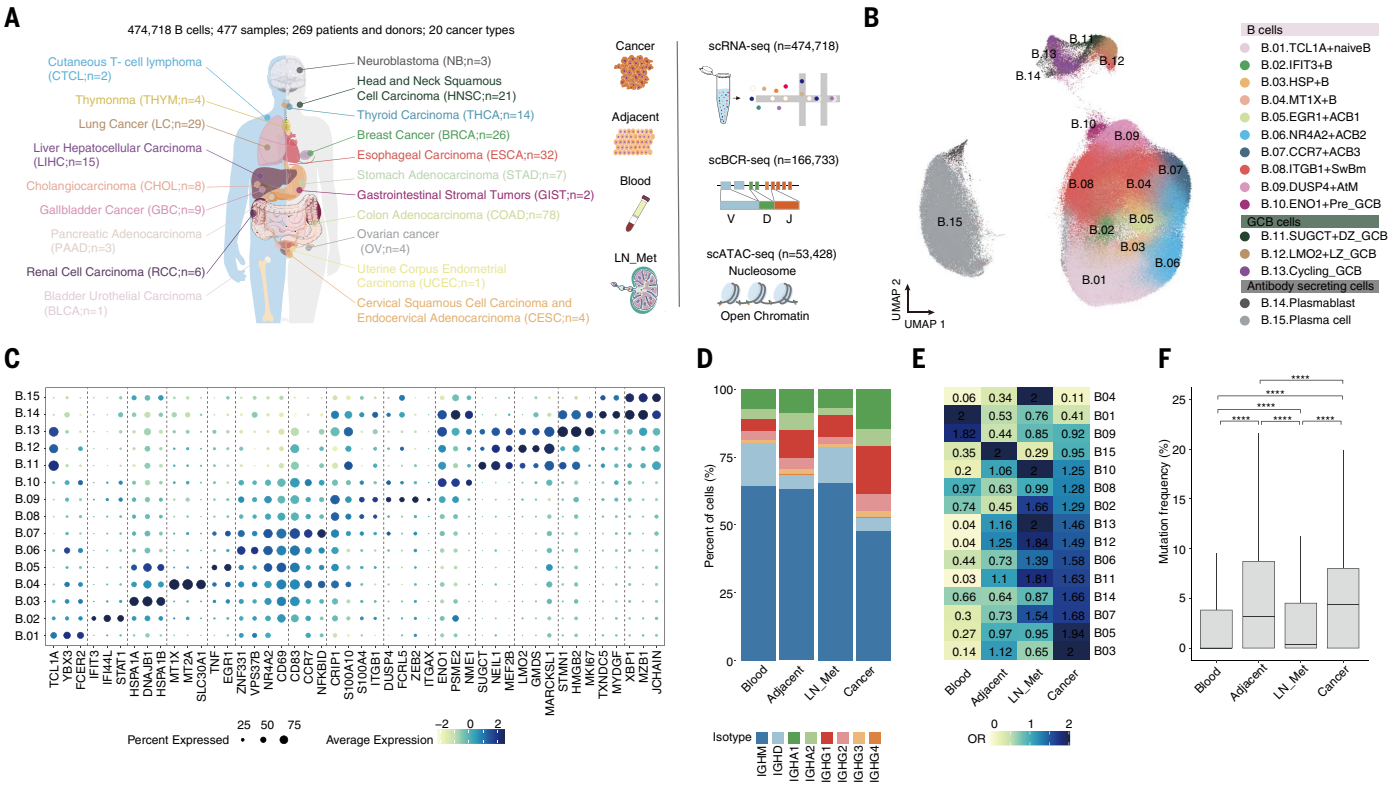


Fig. 1. Single-cell profiling of B cells across different human cancers. (A) Schematics of the pan-cancer single-cell transcriptome, BCR repertoire, and chromatin accessibility of B cells. (B) Uniform manifold approximation and projection (UMAP) visualization of 15 B cell subsets. (C) Dot plot for the expression of marker genes in B cell subsets. Colors represent maximum-normalized mean expression of marker genes, and the size represents the percentage of cells expressing these genes. (D) Stacked bar plots showing the

isotype distribution in the corresponding tissues within total B cells. (E) Heatmap showing the odds ratios (ORs) of B cell subset distribution in each tissue. The OR value represents the preference distribution in corresponding tissue. (F) Box plots of hypermutation frequency on the immunoglobulin H (IgH) chain of total B cells across different tissues. Data indicate median with interquartile range (IQR), and whiskers indicate minimum and maximum measurement. *****P* < 0.0001. *P* values were determined by the two-sided Wilcoxon test.

Overall, our single-cell atlas creates a pan-cancer blueprint for B cell investigation (available at <http://pancancer.cn/B/>).

Transcriptional diversity of B cells

Unsupervised clustering analysis identified 15 distinct B cell subsets, all of which were reproducibly observed across cancer types and exhibited distinct tissue- and cancer-type preferences (Fig. 1B; fig. S1, D and E; and table S2). Among them, typical B cell subsets representing different B cell maturation stages were identified, including one naïve B cell (*TCL1A*, *FCER2*, and *IGHD*), three activated B cells (ACBs; *CD69* and *CD83*), one Bm (*CRIP2* and *ITGB1*), three GC B cells [*LMO2*⁺Light Zone (LZ), *CXCR4*⁺Dark Zone (DZ), and *MKI67*⁺cycling], and two ASCs (including *TXNDC5*⁺PBs and *MZB1*⁺ PCs) (Fig. 1C and fig. S1F).

We next identified four additional subsets that were largely uncharacterized previously (26–30) (fig. S2A). The first was an interferon-stimulated gene-positive naïve B cell subset (B_02) that highly expressed *IFIT3*, *IFI44L*, and *ISG15*, which was associated with injury and was expanded during mucosal healing (31).

The second subset was stressed B cells (B_03), predominant in the tumor with high expression of heat shock proteins, as confirmed by 37°C collagenase and 4°C mechanical dissociation and multiplex immunohistochemistry (miHC) (fig. S2, B and C). In comparison with stressed T cells (32), stressed B cells also resided in the center of TLSs, with high expression of hypoxia-related genes, such as *HIF1A*, *NFE2L2*, *RELB*, *NFKB1*, and *NFKB2* (fig. S2D). The third subset was pre-GC B cells (B_10) that were highly enriched in LN_Mets and tumors as compared with adjacent normal tissues and blood, expressing *PSME2*, *NME1*, and *ENO1*. Pre-GC B cells showed transcriptional similarity to naïve B cells, GC B cells, and PBs, respectively, indicating their transitional stages (Fig. 1C).

The fourth subset we identified was a previously unknown AtM B cell subset (B_09) expressing *DUSP4*, *ITGAX* (*CD11c*), *FCRL5*, *ZEB2*, and *FGR*. Previously documented as ABCs, CD27 IgD[−] B_{DN}, *FCRL4*⁺ B cells, or exhausted Bm cells reported in aging mice, autoimmune diseases, tonsil, and chronic infection, respectively (16, 19, 23, 24), this subset is now demonstrated to widely exist in the TME (Fig. 1C). We

collected and isolated previously published scRNA-seq AtM B cells from autoimmune diseases and hepatitis B virus (HBV) infection datasets (table S1) and performed differentially expressed gene analysis, showing that tumor-infiltrating AtM B cells highly expressed stress-related genes (*FOSB* and *HSPA14*), interferon-induced genes (*IFITM1* and *IFI30*), and activation-related genes (*CD83*, *NR4A1*, and *CD69*), whereas naïve-related genes (*TXNIP* and *IGHD*) and metabolic-related genes (*MT-ATP6* and *NDUFB1*) were enriched in autoimmune diseases and HBV infection (fig. S2, E and F, and table S2). Likewise, tumor AtM B cells were enriched in pathways of inflammatory response and hypoxia, whereas angiogenesis and metabolism pathways were dominant in those from autoimmune diseases and HBV infection (fig. S2G). These data highlight that AtM B cells in TME could be functionally distinct, despite their phenotypic concordance with those from autoimmune diseases and virus infections.

Integrative analysis of tissue distribution and transcriptional and repertoire features highlighted the potential tumor-reactive B cells from different tissue sources. TIBs contained

more IGHG and IGHA than B cells from non-tumor tissues, whereas B cells in blood were enriched with IGHD and IGHM, suggesting that TIBs were antigen-experienced and underwent CSR (Fig. 1D). By the odds ratio (OR) analysis (33), *HSP*⁺ B cells, *EGRI*⁺ACBs, and PBs showed a strong preference in tumor; *MTIX*⁺ B cells, cycling GC B cells, pre-GC B cells, GC_{LZ}, GC_{DZ}, and *IFIT3*⁺ B cells were significantly enriched in LN_Mets; *TCLIA*⁺ naïve B cells and *DUSP4*⁺ AtM B cells were highest in blood; and *MZBI*⁺ PCs were dominant in adjacent normal tissue (Fig. 1E). Again, TIBs had significantly more IGH mutations than those in other tissues, with GC B cells and two ASC subsets exhibiting tumor-specific clonal expansion, indicating their tumor reactivity with strong antigen-driven antibody affinity (Fig. 1F and fig. S2H). In addition, *ITGB1*⁺ Bm and *DUSP4*⁺ AtM B cells showed comparable enrichment in tumor, LN_Mets, and blood, slightly higher than adjacent normal tissue, implying intertissue migration potential. Indeed, the STARTRAC migration (pMigr) index (34), which quantifies the possibility of migration between tissues, and the Jaccard index (35), which illustrates the number of BCR clones shared between different tissues, showed that *ITGB1*⁺ Bm and AtM B cells had moderate mobility between blood and LN_Mets or tumor in about half of the cancers (fig. S2, I to K).

Heterogeneity of tumor-infiltrating antibody-secreting cells

ASCs are terminally differentiated B cells that execute effector function by producing antibodies (36). Compared with the existing data classifying ASCs into a homogenous subset (14, 28, 37, 38), our high-resolution map revealed that tumor-infiltrated ASCs exhibited multifaceted diversities in tissue distribution, cancer-type preference, BCR repertoire, and transcriptomic profile. First, according to the Shannon equitability index (39), the proportion of ASCs showed great variation, with tumor-infiltrated ASCs showing higher diversity than those in adjacent normal tissues and blood ($P < 0.01$), whereas the diversity of non-ASCs was the highest in LN_Mets (fig. S3A). Second, we observed a higher diversity of BCR clonotypes along with an increased number of expanded BCR clonotypes. However, there was a lower clonal frequency observed in tumors, suggesting a higher homogeneity of clone size in tumors than in other tissues, where a small number of expanded clones dominate the repertoire (fig. S3B). Third, cancer types exerted a strong influence on the frequency of ASCs (fig. S1D). Of the two ASC subsets, the median frequency of tumor-infiltrated *MZBI*⁺ PCs was 12.52% of TIBs, ranging from high (70.18%) in colon adenocarcinoma (COAD) to barely detectable in cholangiocarcinoma (CHOL) (1.85%) and stomach adenocarcinoma

(STAD) (3.32%) [analysis of variance (ANOVA), $P < 2e^{-16}$] (fig. S3, C and D). The median frequency of tumor-infiltrated *TXNDC5*⁺ PBs was 0.13% but still varied among different cancer types ($P = 1.1e^{-5}$). Last, ROGUE analysis (40), which quantifies the transcriptomic purity of identified cell types, further confirmed that of the 15 B cell subsets, 2 ASC subsets exhibited the highest heterogeneity of gene expression (fig. S3E). Subclustering of the ASC compartment (removing *TXNDC5*⁺ PBs) revealed 10 distinct PC subclusters with specific gene signatures and tissue distribution (Fig. 2, A and B; fig. S3F; and table S2). Among them, PC04.*HSPA1A*, PC05.*NEAT1*, and PC08.*IFI6* were dominant in the tumor, PC09.*RGS13* and PC10.*SPINK2* accumulated in adjacent normal tissue, and PC02.*RGS13* was prevalent in LN_Mets. In parallel, PC01.*NME2* and PC07.*CD83* showed comparable enrichment in LN_Mets and blood, PC03.*DUSP5* was equally distributed in LN_Mets and tumor, and PC06.*SLC3A2* was similarly distributed in the tumor and adjacent normal tissue (Fig. 2C). These findings may help explain the controversial roles of PCs in patient prognosis (5, 41, 42), which could be ascribed to the variability of different PC subsets and cancer types. Collectively, these data indicate that cancer origin has considerable impact on the frequency of ASCs and also imply different B cell origins for ASCs.

Both extrafollicular and germinal-center paths are hijacked by cancer

Although B cell pathways derived from GC (1, 17, 42) or extrafollicular (EF) responses are commonly observed in autoimmune diseases and chronic infections (21, 43, 44), the potential for the tumor ecosystem to influence distinct B cell evolutionary trajectories has not been thoroughly explored. We thus used gene expression to infer the trajectories of ASC differentiation and observed that ASCs were mainly derived from GC B cells (fig. S3G), which is consistent with previous studies (2, 45). However, given that ASC subsets were significantly distinguished from the other subsets by unsupervised clustering on the basis of high expression of PC markers, traditional transcriptome-based trajectory analysis may not accurately reflect the true differentiation state. To address this, we used a BCR clonal-sharing strategy and unsupervised clustering to identify the progenitors of ASCs with STARTRAC [single T cell analysis by RNA-seq and TCR (T cell receptor) tracking] pairwise transition indices (pTrans) (34), Jaccard indices (35), and somatic hypermutations (SHM), because the BCR repertoire is a reliable molecular tag for tracing the lineage of ASCs. We showed that AtM and Bm cells were the two primary B cells clonally shared with ASCs, suggesting that ASCs may originate from both canonical GC and alternative EF pathways in

the TME (Fig. 2D and fig. S3, H and I). To validate this, tumor-infiltrated naïve B, Bm, and AtM B cells from liver hepatocellular carcinoma (LIHC) patients were sorted and stimulated by using in vitro B cell differentiation methods (24). Indeed, AtM B cells could differentiate into ASCs more efficiently than naïve B cells ($P = 0.035$) but more weakly than Bm cells ($P = 0.0027$) (Fig. 2, E and F). In addition to AtM and Bm cells, we also identified pre-GC, cycling GC, and GC_{LZ} B cells sharing BCRs with PBs, and *IFIT3*⁺ B, *HSP*⁺ B, and *EGRI*⁺ACBs sharing BCRs with PCs, which were validated by phylogenetic lineage trees of shared clones between ASCs and these subsets (Fig. 2G and fig. S3, J to L).

We next evaluated the prevalence of the two paths and classified the ASCs sharing BCRs with canonical Bm (66% of GC-derived ASCs), cycling GC (13%), GC_{LZ} (12%), and GC_{DZ} (7%) as GC-derived ASCs, whereas those ASCs overlapping with other B cell subsets, including AtM (47% of EF-derived ASCs), ACB (29%), *IFIT3*⁺ B (8%), *HSP*⁺ B (6%), pre-GC (5%), and *MTIX*⁺ B (3%) cells, were classified as EF-derived ASCs at the cellular level (fig. S4A). Subsequently, we used label transfer to acquire more EF- and GC-derived ASCs on the basis of the gene expression profile of already classified ASCs and then validated label-transfer accuracy (fig. S4B). With the defined EF-derived and GC-derived ASCs, we further computed the EF index and integrated with SHM to classify patients into two distinct groups—either EF-dominant or GC-dominant at the patient level—and this classification was further validated and refined by pTrans score (fig. S4, C to E). Both paths could be observed in individual patients and cancers despite obvious patient- and cancer-type preference, indicating both GC and EF responses as universal paths for ASC differentiation (Fig. 2H). COAD and thyroid carcinoma (THCA) used GC_{DZ}, GC_{LZ}, and cycling GC B cells, whereas gallbladder cancer (GBC), STAD, lung cancer (LC), and thymoma (THYM) used Bm cells as preferred to the GC pathway. Conversely, the AtM B cell-dominant EF path was enriched in LIHC, pancreatic adenocarcinoma (PAAD), cervical squamous cell carcinoma and endocervical adenocarcinoma (CESC), and head and neck squamous cell carcinoma (HNSC), whereas pre-GC, ACB, and the *IFIT3*⁺-dominant EF path were abundant in renal cell carcinoma (RCC), bladder urothelial carcinoma (BLCA), and gastrointestinal stromal tumor (GIST), respectively (Fig. 2H and fig. S4F). AtM B cells had pTrans scores comparable to those of cycling GC and Bm cells in ovarian cancer (OV) and breast cancer (BRCA), respectively, indicating the equal presence of both pathways in these cancers (fig. S4G). In addition to intersubset BCR sharing, the developmental potential of GC and EF pathways could be validated by the expression

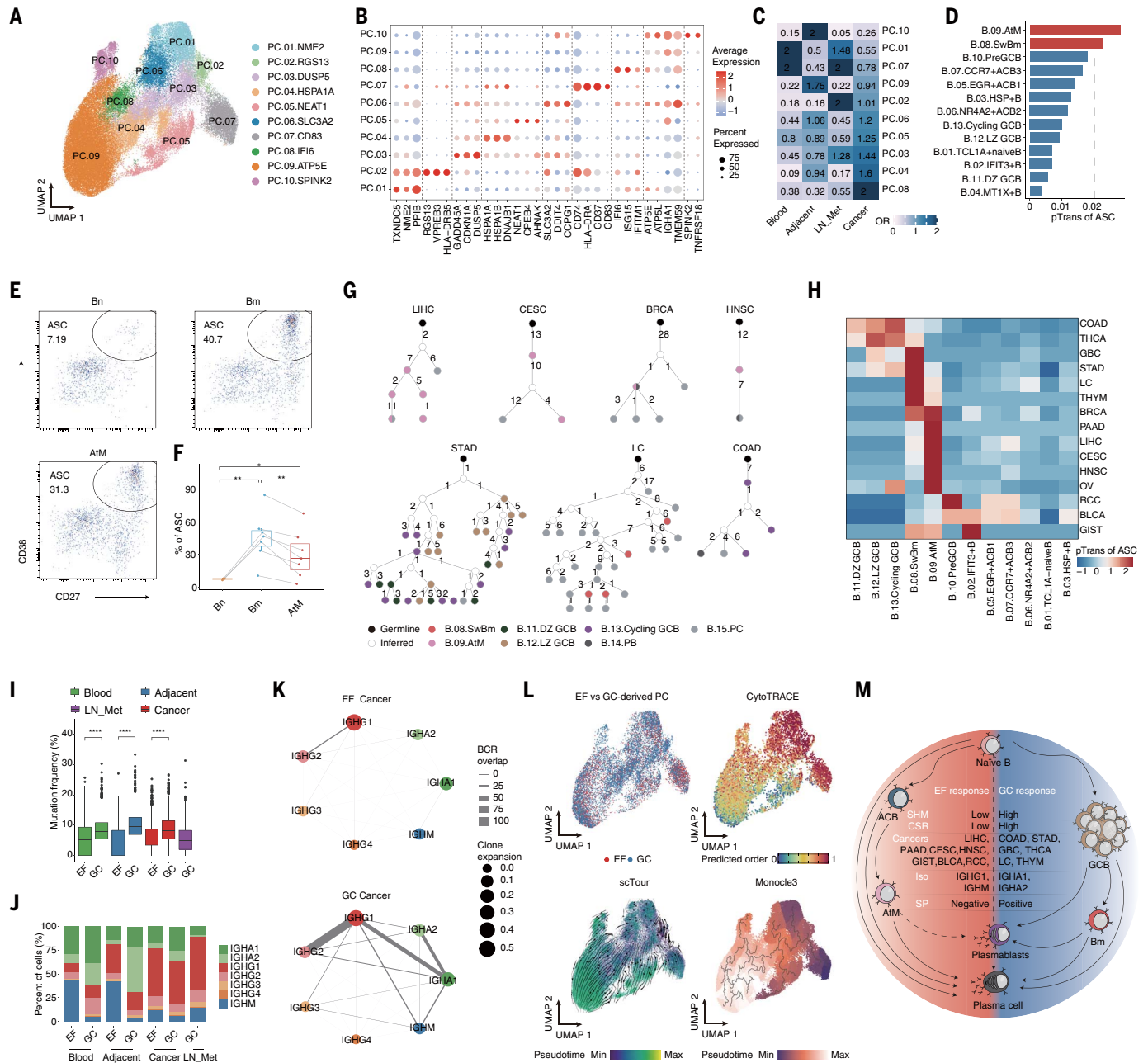


Fig. 2. Two developmental pathways of plasma cells in human cancer. (A) UMAP visualization of 10 PC subsets. (B) Dot plot for expression of marker genes of identified PC subsets. Color represents the maximum-normalized mean expression of cells expressing marker genes, and size represents the percentage of cells expressing these genes. (C) Heatmap showing the ORs of PC subset distribution in each tissue. The OR value represents the preference distribution in corresponding tissue. (D) Pairwise transition index (pTrans) showing BCR overlap between non-ASCs and ASCs. Top two subsets with high connection with ASCs are highlighted. (E and F) The tumor-infiltrated naive (Bn), memory B (Bm), and atypical memory (AtM) B cells from LIHC patients were isolated by FACS and stimulated in vitro with R848, IL-2, IL-10, and IL-21 for 11 days (donor number = 3 to 7). Representative flow plot (E) and frequency (F) of ASCs in CD19⁺ B cells were detected by FACS. Data indicate median with IQR, and whiskers indicate minimum and maximum measurement. (G) (Top) Phylogenetic lineage trees of shared clonotypes between ASC cells and AtM B cells and (bottom) between ASCs and Bm or GC B cells. Acquisition of somatic mutation is shown by a number on the branch indicating the number of mutations and branches. Putative unmutated

common ancestors are indicated by a white circle. (H) Heatmap showing the pTrans between ASCs and other B cell subsets according to different cancer types. Color represents the z-score-scaled pTrans value by row. (I) Boxplots showing the somatic hypermutation (SHM) frequency of EF- and GC-derived ASCs across different tissues. Data indicate median with IQR, and whiskers indicate minimum and maximum measurement. (J) Stacked column charts showing the frequencies of IGH isotypes in EF- and GC-derived ASCs across different tissues. (K) CSR frequencies of EF-derived (up) and GC-derived (down) ASCs in cancer. The thickness of the line indicates the number of shared clonotypes between two Ig isotypes, and the dot size represents the clone expansion of each isotype. (L) UMAP plots showing the identification of EF- and GC-derived ASCs (top left), development order by CytoTRACE (top right), differentiation state inferred by scTour (bottom left), and differentiation state inferred by Monocle3 (bottom right). (M) Sketch map showing the differentiation paths of EF versus GC responses. The middle table summarizes the features of EF and GC in the TME. For (F) and (I), **P* < 0.05, ***P* < 0.01, *****P* < 0.0001. *P* values were determined by the two-sided Wilcoxon test.

signatures of non-ASC subsets in ASCs (fig. S4H). For example, *PCOLNME2* expressed pre-GC signature genes, including *PSME2* and *ENO1*, and these pre-GC B cells shared BCR with both *TXNDC5*⁺ PBs and *MZBI*⁺ PCs in BLCA and RCC. Similarly, *PC04.HSPA1A* expressed *HSPA1A* and *HSPA1B*, and *HSP*⁺ B cells favored to share BCR with *MZBI*⁺ PCs in BLCA, suggesting heterogeneous differentiation pathways toward ASCs. Together, these data suggest that two evolutionary paths (GC and EF) are hijacked by distinct cancer ecosystems.

We next explored the BCR repertoire and matched transcriptional features of these two evolutionary paths. The median frequency of EF-derived ASCs in tumors was 1.52% [0.35 to 4.37% (interquartile range)], which was significantly higher than that of GC-derived ASCs (0.45%; 0.13 to 2.29%) (fig. S5A). By contrast, the diversity of gene expression and BCR clonotypes was reduced in EF-derived ASCs (fig. S5, B and C). These cells were characterized by oligoclonal expansions, with the top five clonotypes accounting for 11.51% of the total repertoire compared with 3.20% in GC-derived ASCs. Indeed, GC-derived ASCs exhibited high diversity and relatively homogeneous clone sizes (fig. S5C). Consistent with previous findings in autoimmune disease (43, 46), EF-derived ASCs in tumors displayed significantly lower SHM than those of GC-derived ones (Fig. 2I and fig. S5D). Isotype analysis demonstrated that EF-derived ASCs contained high interferon-induced immunoglobulin G (IgG) isotype and CSR relative to IGHG1 in tumor, along with higher IGHM enrichment and no CSR occurring in adjacent normal tissue and blood (Fig. 2J). By contrast, GC-derived ASCs showed significantly higher expansion and CSR relative to IGHA1 and IGHA2 in tumor, adjacent normal tissue, and blood, which was linked to tumor antigen-specific recognition and anti-tumor immunity (47) (Fig. 2K and fig. S5E). These differences may stem from the intrinsic distinctions between EF- and GC-derived ASCs, with EF-derived ASCs highly expressing *IRF8*, *CD83*, and *HLA-DR*, exhibiting an earlier immune-response phenotype (48) and enriching for pathways of hypoxia and inflammatory response. Conversely, GC-derived ASCs showed higher expression of immunoglobulin genes and exhibited a more mature PC phenotype—which is consistent with their appearance at later stages of the immune response—and were enriched for several metabolic pathways, including glycolysis, oxidative phosphorylation, and fatty-acid metabolism, highlighting extensive energetic and biosynthetic demands during CSR and GC response (fig. S5, F and G, and table S2). To compare the early and late differentiation stages of EF- and GC-derived ASCs, we used three methods to infer the differentiation state, including cytoTRACE (49), scTour (50), and Monocle3 (51), and showed that EF-

derived ASCs were in the early stages compared with terminal differentiation of GC-derived ASCs, indicating an early EF response and a delayed GC response (52) (Fig. 2L and fig. S5H). These results were further validated because EF-derived ASCs were enriched for *PC07.CD83*, with high expression of *HLA-DR* and *FOXP1*, which has been demonstrated to impede germinal center (GC) formation and repress human PC differentiation (53, 54) (fig. S5I). Additionally, the enrichment of the IGHM isotype in *PC07.CD83* further aligns with the characteristics of early immunoglobulin M (IgM) ASCs in bone marrow (48) (fig. S5, J and K). GC-derived ASCs were enriched for *PC03.DUSP5*, *PC04.HSPA1A*, and *PC10.SPINK2*. LIHC-infiltrated ASCs exhibited significantly higher EF-derived ASC markers, such as *CD20*, *HLA-DR*, *CD37*, *CD74*, *CD83*, and *IRF8*, than did COAD ones (fig. S5, L and M). These results were also validated with mIHC staining for *IRF8* and *HLA-DR* in ASCs, using tissue microarray (TMA) encompassing both EF-dominant (BLCA, CESC, GIST, LIHC, PAAD, RCC; *n* = 49) and GC-dominant cancers (COAD, STAD, and LC; *n* = 23) (fig. S5N). The frequency of *IRF8*⁺ ASCs (median in EF = 13.67%, median in GC = 6.58%; *P* = 0.0315) and *HLA-DR*⁺ ASCs (median in EF = 50%, median in GC = 34.09%; *P* = 0.0066) among *CD79*⁺*CD20*[−] ASCs was significantly higher in EF-dominant cancers compared with GC-dominant cancers (fig. S5O). Taken together, these results further support the conception that the early response of EF-derived ASCs contrasts with the delayed response of GC-derived ASCs.

GC-derived ASCs had more frequent mutations in activation-induced cytidine deaminase (AID) WRC/GYW and WA/TW hotspots than did EF-derived ASCs (55, 56) (fig. S5, P and Q). Specifically, the mutation frequency of GC-derived ASCs in the AID hotspots was 6.05% (4.37 to 8.44%) for IGHA and 4.61% (1.05 to 7.07%) for IGHM. These frequencies were significantly higher than those in EF-derived ASCs, which were 4.65% (2.63 to 6.95%) for IGHA (*P* = 2.5e^{−12}) and 0% (0 to 2.93%) for IGHM (*P* < 2.2e^{−16}). GC-derived ASCs displayed positive selection for replacement mutation in the complementarity-determining region (CDR) of IGHA2 and IGHG2, further supporting their increased CSR (fig. S5R). By contrast, IGHG4 was increased in EF-derived ASCs compared with GC-derived ASCs, which was consistent with a recent report showing IgG4-related autoimmune disease with enriched EF response (57). Likewise, comparing *IGHV* gene usages between EF- and GC-derived ASCs in tumors revealed that GC-derived ASCs overexpressed *IGHV1-8* and *IGHV1-24* (fig. S5S), where *IGHV1-8* preference was correlated with higher SHM rate (58), indicating their strong antigen-specific affinity. EF-derived ASCs up-regulated a group of *IGHV* genes (*IGHV3-15*,

IGHV3-23, *IGHV3-33*, *IGHV3-49*, *IGHV4-34*, and *IGHV4-59*), and increased usage of these *IGHV* genes has been associated with COVID-19, viral infections, and autoimmune disease (43, 46, 59, 60), implying a polyreactive BCR repertoire. Together, our data indicate that distinct cancer ecosystems hijack two evolutionary paths (EF and GC) and reveal differences in BCR repertoires and transcriptional features between EF- and GC-derived ASCs, highlighting their distinctive roles in antitumor response (Fig. 2M).

Regulatory elements epigenetically license extrafollicular and germinal-center balance

Transcription regulation is fundamental for dictating and maintaining B cell identity (61–64), but the tumor-specific regulatory network and the molecular drivers that contribute to TIBs' identity remain poorly defined. To address this challenge, we applied paired scRNA-seq and scATAC-seq (ATAC, assay for transposase-accessible chromatin sequencing) to characterize the epigenomics of TIBs and obtained chromatin accessibility maps for 53,428 TIBs from nine patients of five cancer types, including GC pathway-dominant cancers (COAD, THCA, LC, and STAD) and EF pathway-dominant cancers (LIHC), with a median of 11,853 distinctly aligned fragments, and ~67.87% of Tn5 insertions within ATAC-seq peaks (Fig. 3A and fig. S6, A to C). By integrating scATAC-seq and scRNA-seq data, we observed eight analogous B cell subpopulations, covering naïve, two ACB, Bm, AtM, GC (DZ and LZ), and PC subsets (Fig. 3A and table S3). As a result, there was strong consistency between subset identities and subset-specific markers annotated in both scRNA-seq and scATAC-seq (Fig. 3B and fig. S6D). Open chromatin accessibility of signature genes for B cell subsets, such as *TCL1A*, *RGS13*, *ITGAX*, and *XBPI*, further confirmed their cellular identities (Fig. 3C and fig. S6E). To define the transcription factors (TFs) involved in modifying chromatin accessibility of different B cell subsets, we identified TFs whose chromatin accessibility activity was significantly and positively correlated with their gene expression. Such analysis generated a full spectrum of enriched TFs, including known B cell lineage TFs such as *PAX5*, *EBF1*, and *SPIB* (65) (fig. S6, F and G). By combining the footprint and signature of TFs (66), we identified shared and distinctive regulatory patterns across B cell subsets (Fig. 3D). For example, regulatory TF activities harbored shared patterns between similar cell states, such as those active in naïve, ACB, and Bm (i.e., *BCL11A*, *ELF2*, *SPIB*, and *STAT6*), or in GC_{DZ} and GC_{LZ} B cells (i.e., *CTCF*, *EBF1*, *HMGAI*, and *POU2F1*) (Fig. 3D). Conversely, a broad spectrum of TFs was specifically enriched in certain B cell subsets (i.e., *ETS1* in naïve B; *TBX21* and *FOSL2* in AtM; *POU2F1*, *MEF2B*, and

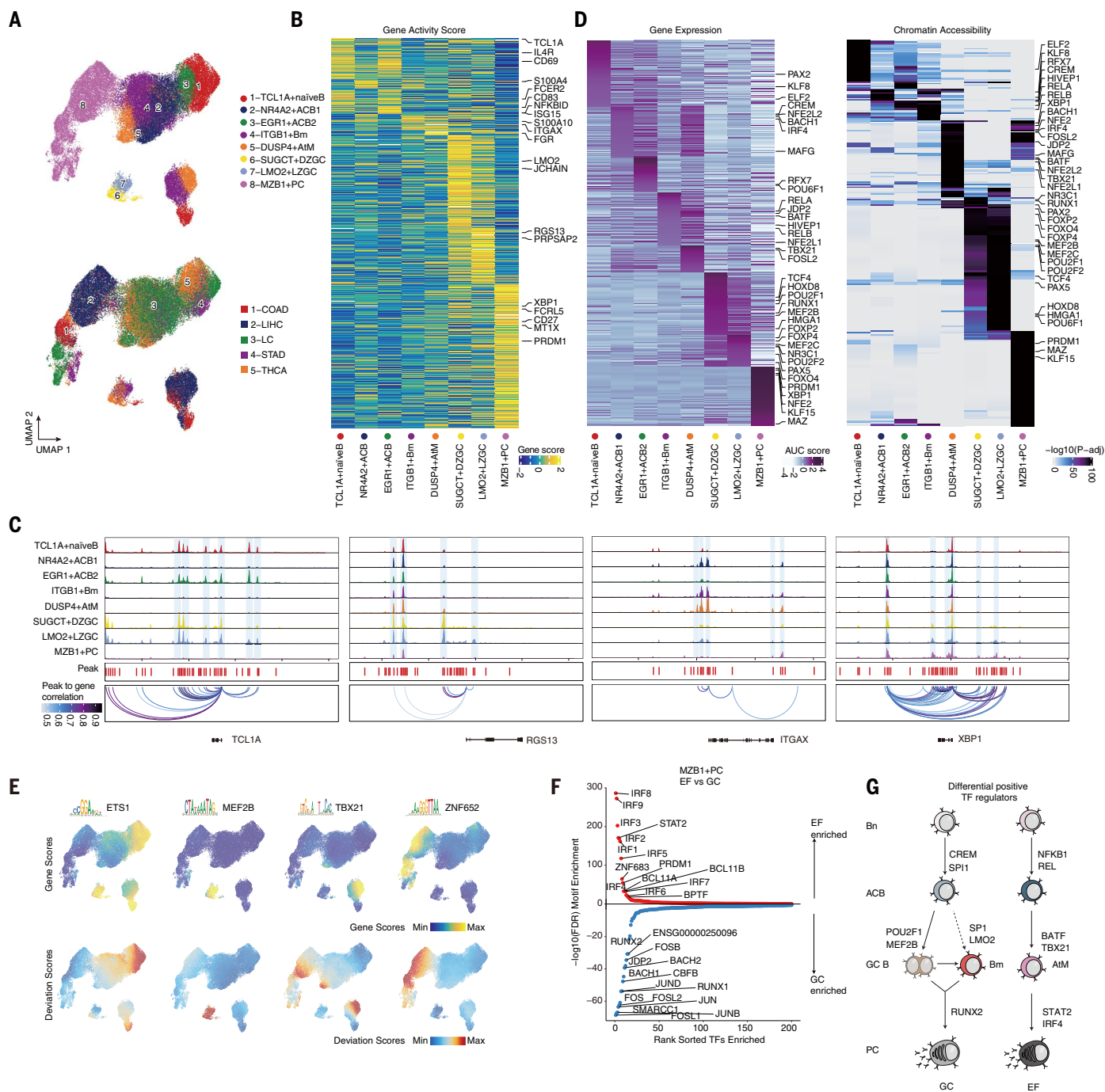


Fig. 3. Single-cell epigenomic profiles of EF and GC responses in human cancer. (A) UMAP visualization of scATAC-seq-annotated B cell subsets (top) and cancer types (bottom). (B) Heatmap of gene activity score determined by scATAC-seq of all marker genes from each subset. Color represents z-score-scaled gene expression. (C) Genome accessibility tracks of indicated signature gene loci in B cell subsets. (D) Heatmap of TF activity by SCENIC (left) and TF motif enrichment at the specific open chromatin regions (right) of the annotated B cell subsets. (E) UMAP visualization of RNA expression (top) and motif

deviation scores (bottom) for B cell subsets' specific TFs. (F) Differentially expressed TF deviation score between EF-dominant and GC-dominant cancer-derived MZB1⁺ PCs were calculated [$\log_2$ fold change (FC) >0.5, Benjamini-Hochberg adjusted P value <0.05, two-sided Wilcoxon test]. Ranking of TF deviation score by P -adjusted value within (red) EF-dominant versus (blue) GC-dominant cancer-derived MZB1⁺ PCs. (G) Schematic of differential putative regulatory TFs driving B cell differentiation from naïve B cells to PCs in (left) GC-dominant cancer and (right) EF-dominant cancer.

BCL6 in GC B; and *IRF2/3/4/5/6/7/8*, *PRDM1*, and *ZNF652* in PCs (Fig. 3, D and E, and fig. S6H). Overall, our data not only unbiasedly shed light on both known and uncharac-

terized TFs in controlling B cell identity but also provide a comprehensive resource of gene regulatory networks for major B cell subsets across cancers.

Upon closer inspection of Bm, AtM, and PCs, we found that they were predominantly composed of cells from specific EF and GC cancer types, prompting us to investigate potential

cancer entity-specific differences in the chromatin landscape of Bm, AtM, and PCs (Fig. 3A). We therefore studied the lineage trajectories of B cells from EF- and GC-dominant cancers (fig. S7, A to C). We identified sequential activities of positive TF regulators, including *PAX5*, *BCL11A*, *POU2F2*, *EBF1*, *REL*, *TBX21*, *PRDM1*, *IRF4*, *RORA*, *RUNX2*, and *XBPI*, which promote B cell-lineage specification, maintenance, CSR, and differentiation (62, 64, 65, 67). These activities varied between EF- and GC-dominant cancers along B cell differentiation. Subsequently, we evaluated the TF deviation scores to pinpoint the differential positive TF regulators between EF- and GC-dominant cancers. Bm and AtM in EF-dominant cancers were enriched with AP-1 factors, including *FOS* and *JUN*, which are involved in BCR signaling pathway and signify B cell activation and differentiation (68) (fig. S7, D and E). By contrast, Bm and AtM in GC-dominant cancers were enriched with nuclear factor κ B (NF- κ B) subunits, including *REL*, *RELA*, and *RELB*, which is consistent with their reported functions in GC B cell maintenance and homeostasis (69). Compared with PCs in GC-dominant cancers that

showed a high *RUNX1-3* deviation score, PCs in EF-dominant cancers showed high activity of TFs that regulate rapid induction of interferon- β (IFN- β) and inflammatory TME, including *IRF1/3/4/5/8* and *STAT2* (70) (Fig. 3F). To validate this, we confirmed that IRF4 and RUNX2 were enriched in tumor-infiltrated ASCs as compared with other B cells and that IRF4 was up-regulated in LIHC-associated ASCs as compared with COAD-associated ASCs, whereas RUNX2 showed the opposite trend (fig. S7, F and G). These results were substantiated by mIHC staining for IRF4 and RUNX2 in ASCs from a TMA including both EF-dominant ($n = 49$) and GC-dominant cancers ($n = 23$) (fig. S7H). The frequency of IRF4⁺ (median in EF = 55.09%, median in GC = 50%; $P = 0.0416$) in CD79a⁺CD20⁺ ASCs was significantly higher in EF-dominant cancers than in GC-dominant cancers, whereas the frequency of RUNX2⁺ ASCs showed the opposite trend (median in EF = 26.11%, median in GC = 56.89%; $P < 0.0001$) (fig. S7I). Collectively, these data highlight a dynamic epigenetic regulatory network that functions in fine-tuning B cell differentiation and selection within cancer ecosystems,

and they underscore the fundamental role of lineage TFs in managing the balance between EF and GC pathways (Fig. 3G).

Extrafollicular AtM B cells harbor an exhaustion phenotype, developing independently of germinal centers

Recognizing the critical role of AtM B cells as primary progenitors for the EF pathway in the TME, we comprehensively examined their phenotype and function. Compared with the previously defined signature markers in non-tumor contexts (21, 22, 24), tumor-infiltrating AtM B cells also showed reduced expression of *CD27*, *CD38*, *IRF8*, and *SDCI*, together with increased expression of *PRDM1*, *IRF4*, and *XBPI* in comparison with Bm cells, the well-characterized progenitors for GC-derived ASCs (1). *XBPI* is known to dampen B cell proliferation and promote differentiation toward ASCs (71), supporting the differentiation potential of AtM to ASCs (Fig. 4A). AtM B cells exhibited an exhausted and resident memory phenotype, distinguished from the effector phenotype of Bm. In brief, AtM B cells up-regulated immuno-checkpoints (*PDCD1*, *CD274*, *CTLA4*, *ENTPD1*,

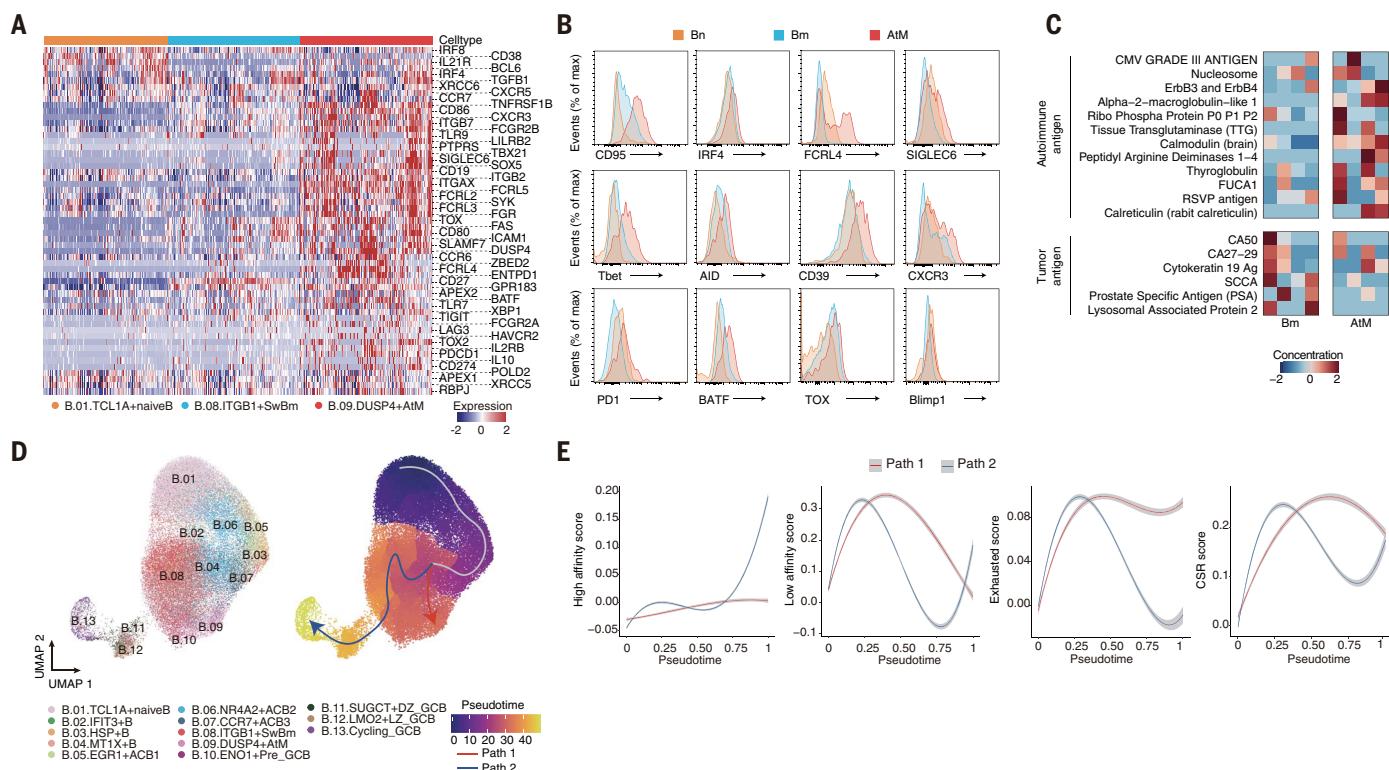


Fig. 4. GC-independent development of AtM B cells in human cancer.

(A) Heatmap showing the relative expression of representative genes in naive, Bm, and AtM B cells. The color scale represents the z-score-scaled gene expression value by row, and each column represents an individual patient. (B) Representative markers of tumor-infiltrated naive, Bm, and AtM B cells from LIHC patient were detected by FACS ($n = 3$). (C) Heatmap showing the detection of autoantigens' and tumor-associated antigens' specific IgA in the supernatant of tumor-infiltrated naive, Bm, and AtM B cells from LIHC patients

after 11 days in vitro stimulation ($n = 4$). (D) (Left) Monocle3 trajectory analysis depicting the developmental trajectories of non-ASC, revealing (right) two major divergent trajectories (from gray skeleton line: red, path 1; blue, path 2). (Bottom right) Cells are color coded for their corresponding pseudotime. (E) Two-dimensional plots showing the dynamic expression scores for high-affinity, low-affinity, exhaustion, and CSR signatures in cells of path 1 (red) and path 2 (blue), respectively, along the inferred pseudotime. The center line indicates linear fit, and shaded lines indicate 95% confidence interval.

LAG3, and *HAVCR2*), exhaustion-associated TFs (*TOX*, *TOX2*, *ZBED2*, *BATF*, *RBPJ*, and *VDR*) (33, 72), and Fc receptor family involved in immunoglobulins and immune complex binding (*FCRL2/3/4/5* and *CD32A/B*) (Fig. 4A). Among them, *FCRL4* is linked to tissue-resident memory cells that could inhibit BCR signaling (16). *CD32B* stands out as the low-affinity inhibitory receptor that inhibits antibody-dependent cellular cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), and B cell activation (73). Similarly to exhausted T cells with overactivated TCR signaling, AtM B cells also highly expressed genes of BCR signaling (*SYK*), immunomodulation, and activation (*TLR7/9*, *CD80*, *CD86*, and *CD72*). These distinctive phenotypes were further corroborated by flow cytometry (Fig. 4B and fig. S8A). These data collectively demonstrate the distinctive immunophenotypes and molecular features of extrafollicular AtM B cells.

To further explore the immune repertoire of extrafollicular AtM B cells, isotype analysis indicated that tumor-infiltrating AtM B cells were less switched and contained higher IGHD and IGHM than did Bm cells ($P < 0.01$), whereas IGHA1, IGHA2, and IGHG2 were prevalent in Bm ($P < 0.01$) (fig. S8, B and C). Moreover, supernatants of tumor AtM B cells from LIHC patients, collected after 11 days of in vitro activation and differentiation, contained lower IgG, IgA, and IgM than did Bm (fig. S8D). Again, antigen microarrays testing the reactivity of these supernatant antibodies revealed that ASCs derived from AtM B cells produced higher levels of autoantibodies, whereas Bm-derived ASCs produced higher levels of tumor-associated antigen-reactive antibodies (Fig. 4C) for all three types: IgG, IgA, and IgM (fig. S8E). In line with these findings, we observed high SHM in Bm but low or medium levels in AtM B cells (fig. S8F). Collectively, our data show that exhausted AtM B cells exhibit impaired antibody production capacity, bystander activation, and weak tumor reactivity.

Another crucial issue is whether AtM B cells could enter GC or originate from bona fide GC B cells (19, 74–76). To address this, we jointly analyzed the BCR repertoire, developmental trajectory, epigenomic regulation, and gene expression features. We observed comparable IGHV usage, CDR3 amino acid length, and SHM between AtM and three GC B cells. These data supported the notion that AtM and GC B clones were generally independent rather than interchangeable (fig. S8, G and H). We thus reconstructed the lineage tree for the most expanded clones of AtM and GC B cells, showing that clonal lineages were predominantly derived from either AtM or GC B cells (fig. S8I). Pseudotime trajectory analysis confirmed that AtM and GC B cells were completely separated on two independent branches, with naïve B cells at the beginning and AtM

(path 1) plus cycling GC B cells (path 2) in the terminal branches (Fig. 4D). Along the trajectory, high-affinity signature score and *CD24*, *CD38*, and *SELL* gradually up-regulated in path 2, whereas low affinity and exhaustion-associated signature score-related genes (i.e., *PDCDI*, *ENTPDI*, and *HAVCR2*) correspondingly up-regulated in path 1, recapitulating B cell development from the naïve state through the activated state and finally to the AtM state (Fig. 4E and fig. S8, J and K). During this process, the degree of exhaustion progressively escalated. This developmental trajectory was uniformly observed in different cancer types (fig. S9A) and was confirmed by scATAC-seq data (fig. S9B).

We further identified the dynamic expression, chromatin activity, and footprint of AtM. Compared with Bm and GC B cells, AtM B cells exhibited high activity of TFs involved in CSR and exhausted phenotype, and they enhanced immune function of isotype-specific IgG⁺ Bm (33, 67, 77, 78), including *TBX21* and *BATF* (fig. S9, C and D), indicating distinct regulatory programs (Fig. 3D). CSR-related signature score and genes (*APEX1*, *APEX2*, *XRCC5*, *XRCC6*, *POLD2*, and *AICDA*) progressively up-regulated along the trajectory, suggesting that AtM B cells may undergo CSR without experiencing GC response (Fig. 4E and fig. S8K). Similarly, AtM had shared BCR with *IFIT3*⁺ B in EF-dominant cancers, including HNSC, OV, GIST, BRCA and CESC, suggesting that naïve B cells induced by interferons could directionally differentiate into AtM without entering the GC (fig. S9, E to G). Collectively, these findings suggest that AtM and GC B cells underwent separate developmental processes leading to EF- and GC-derived ASCs.

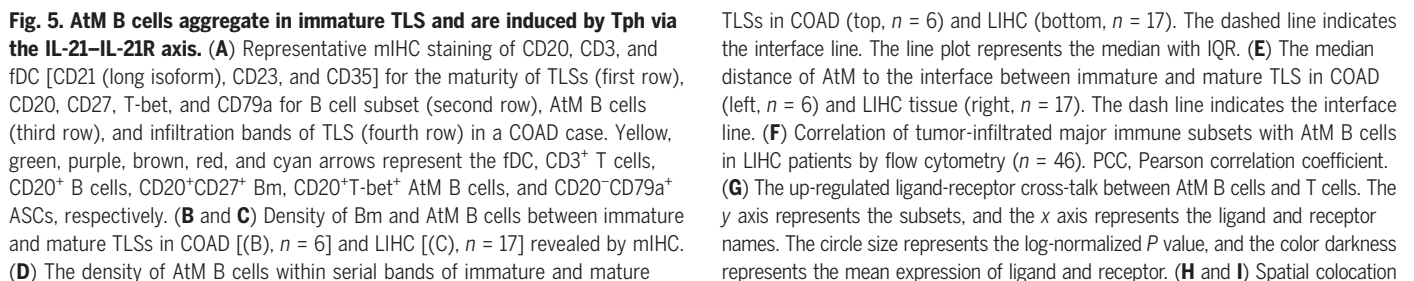
Extrafollicular B cells spatially reside in immature tertiary lymphoid structures

TLSs are often formed by B cells with varying sizes and cellular compositions. However, the spatial architecture and relationship between TLSs and TIBs, specifically AtM B cells, remain unclear. We found that AtM B cells highly expressed TLS signature (7) and were confirmed in pan-cancer TCGA data, indicating their potential localization in TLSs (fig. S10, A and B). We thus performed mIHC staining of TMA containing EF-dominant cancers [LIHC ($n = 358$; $n = 180$) and PAAD ($n = 48$)] and GC-dominant cancers [COAD ($n = 98$), STAD ($n = 52$), and LC ($n = 90$)] to characterize TLSs and AtM B cells (table S1). By using our well-established TLS quantification pipeline (8), we divided tumor tissue into TLS and non-TLS regions and observed a significant correlation between AtM B cells and TLSs, suggesting a potential role for AtM B cells in TLS formation (fig. S10, C and D).

Given the clinical significance of TLS maturation [mature TLSs: stage III and IV GC-like structure containing follicular dendritic cells

(fDCs); immature TLSs: stage I and II lymphoid aggregates] and the rarity of GC-like TLS in TME (8, 79–82), we further screened whole-tissue sections with different TLS stages from LIHC ($n = 17$) and COAD ($n = 6$). TLSs were classified into four stages on the basis of the cell number of T and B cells together with fDC staining (8) (Fig. 5A). AtM B cells were significantly enriched in immature TLSs, whereas Bm was mainly located in mature TLSs in both COAD and LIHC (Fig. 5, B and C). No difference in ASC density was observed across different TLS stages (fig. S10E). As expected, AtM B cells were predominantly enriched in EF-dominant cancers, such as GIST, CESC, LIHC, and HNSC, whereas GC B cells were highly abundant in GC-dominant cancers, such as COAD, THCA, and LC (fig. S10F), indicating that GC-dominant cancers mainly contained mature TLSs versus immature ones in EF-dominant cancers. Cancer with equal presence of EF and GC paths, such as OV, showed moderate infiltration of AtM and GC_{LT} B cells as compared with other cancers. Spatially, we calculated the density of AtM B cells within 200 μ m of the inner and outer edges of TLSs or follicles and the distance of AtM B cells to the interface. We observed that AtM B cells were predominantly located in the center of immature TLSs, which is consistent with early EF response and facilitation of TLS formation (Fig. 5, D and E, and fig. S10, G and H). As TLS matured, AtM B cells migrated to the periphery, while naïve B cells consistently resided in the center of TLSs (fig. S10C). These data validate the GC-independent development of AtM B cells and also suggest that TLS maturation stages drive distinct B cell differentiation in EF- and GC-dominant cancers.

This particular location of AtM B cells was echoed by the up-regulation of homing receptors such as CCR7, CXCR3, and EBI2 (encoding *GPR183*) (Fig. 4A), which could drive AtM B cells into the B-zone-T-zone boundary to await T cell help (52, 83). Indeed, the strong correlation between AtM B cells and T follicular helper (Tfh) or type 1 T helper (Th1) cells, which secrete IFN γ , IL-21, and CD40L in viral infection (76, 84), points to the possibility of cross-talk between these T cell types and AtM B cell development. Flow cytometry and pan-cancer TCGA data deconvoluted by BayesPrism (85), a Bayesian statistical model that infers cell type composition using our paired CD45⁺ scRNA-seq data as reference, showed that AtM B cells positively correlated with PD1^{Hi} exhausted T cells (Fig. 5F and fig. S11, A to D), especially PD1^{Hi}CD4⁺ T cells exhibiting a PD1^{Hi}CXCL13⁺CXCR5⁺ phenotype (named as peripheral helper T cells, Tph) that can promote TLS formation and B cell terminal differentiation in autoimmune diseases (86, 87). Ligand-receptor analysis provided additional evidence supporting the cross-talk between



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correlation (H) and representative images (I) between AtM B cell signature and TLS or PD1^{Hi}CD4⁺ Tph in 81 spatial transcriptomics datasets from nine cancer types. Correlation was computed by Spearman rho. Data sources and accessions are summarized in table S4. (J) Representative mIHC staining showing colocation of PD1^{Hi}CD4⁺ Tph and AtM B cells in LIHC tumor tissue. Green, yellow, red, and cyan arrows represent the PD1^{Hi}CD4⁺ Tph, CD20⁺T-bet⁺ AtM B cells, and CD20⁺CD138⁺ PCs, respectively. (K) (Left) Infiltrating density and (right) frequency of PD1^{Hi}CD4⁺ Tph cells within a 20- μ m radius surrounding AtM B cells in TLS and non-TLS regions of the tumor and adjacent areas from LIHC TMA

(*n* = 180). (L and M) Tumor-infiltrated T_{reg}, Th, and PD1^{Hi}CD4⁺ Tph cells were sorted and co-cultured with healthy blood B cells at ratio of 1:5 in the presence or absence of isotype and anti-IL-21R for 7 days. Representative flow plots (L) and frequencies (M) of AtM B cells were determined by FACS (*n* = 6 to 8). Data indicate median with IQR in (B) to (D), (E), and (K), and whiskers indicate minimum and maximum measurement in (H) and (M). ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001; ns, no significance. *P* values were determined by the two-sided unpaired Wilcoxon test for (B) and (C), the two-sided paired Wilcoxon test for (K), and the two-sided paired Student's *t* test for (M).

PD1^{Hi}CD4⁺ Tph cells and AtM B cells, particularly through the IL-21R–IL-21 axis, as compared with Bm cells. This interaction implies a potential role for PD1^{Hi}CD4⁺ Tph cells in promoting AtM B cell differentiation (Fig. 5G and fig. S11E), which was also validated by another independent algorithm, CellChat (88) (fig. S11F). Visualizing CD4⁺PD1^{Hi} Tph and AtM B cells through published spatial transcriptomics datasets [81 datasets from nine cancer types, including BRCA, CESC, COAD, LIHC, LGG, OV, prostate adenocarcinoma (PRAD), RCC, and skin cutaneous melanoma (SKCM)] revealed a positive correlation between them in TLSs (Fig. 5, H and I, and table S4), which was further supported by mIHC staining on LIHC TMA (*n* = 180), showing significantly higher CD4⁺PD1^{Hi} Tph cell density and frequency in close proximity to AtM B cells in TLS than in non-TLS regions of both tumor and adjacent tissues (Fig. 5, J and K). To validate this, LIHC-infiltrated PD1^{Hi}CD4⁺ Tph, Th, and regulatory T (T_{reg}) cells were sorted and co-cultured with either peripheral total B or sorted naïve B cells from healthy donors (Fig. 5L and fig. S11G). The results showed that Tph cells induced differentiation of AtM B cells more efficiently than Th and T_{reg} cells, which was significantly attenuated by IL-21R blockade (Fig. 5M and fig. S11H). Together, these data suggest that tumor-infiltrated Tph cells significantly contribute to AtM B cell differentiation.

Glutamine metabolism establishes the epigenetic identity of extrafollicular AtM B cells

We next examined what factors mechanistically drive the primary response toward EF versus GC paths in TME. Pathway analysis of B cells revealed a strong variance of metabolic pathways across human cancer types, suggesting the potential metabolic regulation of B cell identity (fig. S12A). We computed the metabolic pathway scores of Bm and AtM B cells using scMetabolism (89) and unsupervised clustering of metabolism-associated genes (Fig. 6A). Glutamine, glutamate, and sphingolipid metabolism were up-regulated in tumor-infiltrated AtM B cells, whereas arachidonic acid metabolism was significantly up-regulated in Bm cells. The core genes in these pathways, such as glutaminase (*GLS*), *GLA*, and *PTGES3*, showed similar trends (fig. S12B). The results

were observed in individual types of cancer without cancer-type preference (fig. S12C). Given the essential role and large abundance of glutamine in TME (90–92), we investigated the association between glutamine and B cell immunophenotypes in our cohort. Through mass spectrometry-based untargeted metabolomics, our data indicated that EF response-dominant cancers (LIHC, CESC, and RCC) have significantly higher glutamine than those in GC-dominant cancers (STAD and LC) (fig. S12D). Moreover, glutamine was significantly higher in tumors than in adjacent normal tissues in LIHC (*n* = 109) (fig. S12E), whereas COAD and STAD, two GC-dominant cancers, exhibited lower glutamine levels than adjacent normal tissues (93). Thus, we hypothesize that glutamine metabolism may have a potential role in priming EF response in TME.

To test this hypothesis, we collected B cells from healthy blood and stimulated them with R848 (a TLR7 agonist), IFN γ , and glutamine. Similarly to IFN γ 's role in the autoimmune disease (21, 22), glutamine significantly induced AtM differentiation compared with R848 alone, whereas GLS inhibition (CB839) significantly attenuated it (Fig. 6, B and C). As control, Bm cells were significantly reduced in response to IFN γ and glutamine stimulation, suggesting the opposing regulatory mechanisms for AtM and Bm cells by glutamine. Moreover, AtM gradually increased and peaked around day 4 after in vitro glutamine stimulation in a dose-dependent manner that maximized at around 10-fold of physiological concentration (fig. S12F). We monitored the oxygen consumption rate and adenosine triphosphate (ATP) production of B cells under glutamine stimulation and observed a global reduction of mitochondrial respiration plus glycolysis (Fig. 6D). This indicates that glutamine may remodel the metabolic traits of B cells, leading us to hypothesize that glutamine suppresses the clonal expansion or positive selection of B cells, thus shifting B cell differentiation toward AtM B cells (94).

We then analyzed the intracellular metabolites of peripheral B cells after R848 and glutamine stimulation and found increased levels of glutamate, α -ketoglutarate (α -KG), and a high ratio of glutamine to glutamate as compared with R848 stimulation alone (fig. S12G).

CB839 treatment inhibited this trend, suggesting that GLS plays an essential role in B cell glutaminolysis. The metabolic fate of ¹³C-labeled glutamine after R848 treatment confirmed that glutamine was catalyzed into glutamate (M5 labeled) and entered into the tricarboxylic acid (TCA) cycle through α -KG (M5 labeled), succinate (M4 labeled), fumarate (M4 labeled), pyruvate (M3 labeled), and lactate (M3 labeled), which is consistent with the reported glutaminolysis route (95) (Fig. 6E). Given that glutamine-derived α -KG could regulate *T-bet* expression in Th1 cells (90, 97), we next asked whether α -KG was also critical for AtM differentiation by supplementing with the cell-permeable α -KG analog dimethyl- α -KG (DM α KG). Indeed, AtM differentiation in GLS-inhibited cells was restored when supplemented with DM α KG, comparable to direct stimulation with glutamine. In addition, direct supplementation of DM α KG with R848 induced a similar level of AtM differentiation, suggesting that glutamine-derived α -KG can directly regulate AtM differentiation (Fig. 6F).

Considering the impact of α -KG on histone methylation and chromatin accessibility remodeling (97, 96), we next questioned its role in establishing the epigenetic identity of extrafollicular AtM B cells. We found that glutamine stimulation led to increased H3K27 trimethylation, whereas CB839 had no effect on H3K27 itself (Fig. 6C). Further ATAC-seq and RNA-seq revealed that glutamine-treated B cells showed robust activation of AtM identity peaks (Fig. 6G), indicating their acquired extrafollicular identity. Consistently, we also observed enriched TF motifs for AtM cell differentiation, including *TBX21* and *BATF* (fig. S12H). We further used glutamine to stimulate B cells with *T-bet* overexpression and observed a significant increase in AtM B cells in the overexpression group as compared with the control, providing additional support for the notion that glutamine induces AtM differentiation through the regulation of *TBX21* (fig. S12, I and J). Pathway enrichment analysis again demonstrated that EF-associated PI3K-AKT-mTORC1 and xenobiotic-metabolism pathways were activated, which is consistent with our scRNA-seq data (mTORC1, mammalian target of rapamycin complex 1) (Fig. 6H). Several regulators of the mTORC1 pathway were also

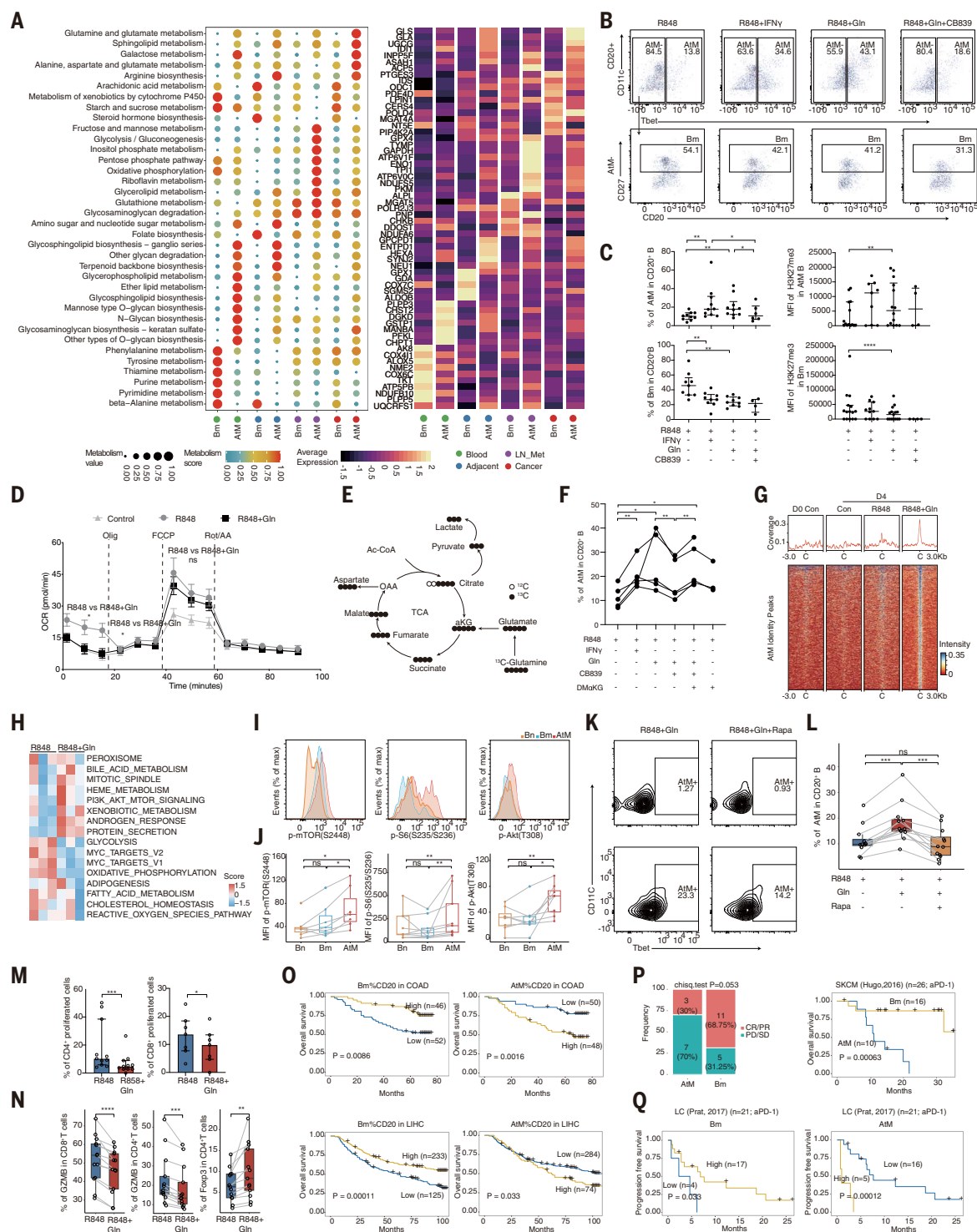


Fig. 6. Glutamine promotes ATM B cell differentiation to acquire immunoregulatory function. (A) Heatmap of median metabolic pathway score (left) and average metabolic gene expression of Bm and AtM B cells across different tissue (right). Both the circle size and color of the dot plot represent the scaled metabolic score (left). The color of the heatmap represents the average metabolic gene expression (right). (B and C) Healthy blood B cells were stimulated with R848 alone or with R848 in the presence of IFN γ or glutamine or glutamine and CB839 for 4 days. (B) Representative flow plots of AtM (top) and Bm (bottom) cells were detected by FACS. (C) (Left) The frequency of Bm and AtM in CD20⁺ B cells and (right) the frequency of tri-methyl-histone H3

(K27) in Bm or AtM B cells were detected by FACS ($n = 6$ to 10). (D) The real-time oxygen consumption rate (OCR) in healthy blood B cells stimulated with phosphate-buffered saline (PBS) (control) or with R848 in presence and absence of glutamine for 24 hours and following the additions of oligomycin, FCCP, and rotenone+antimycin A (Rot/AA) ($n = 4$). The plot is mean $\pm$ SD. (E) Metabolic tracing analysis of ¹³C-labeled glutamine in healthy blood B cells stimulated with R848 for 24 hours ($n = 4$). Indicated labeled metabolites in R848 and glutamine-stimulated B cells were detected by MS. (F) Healthy blood B cells were stimulated with R848 alone or with R848 in the presence of IFN γ or glutamine; or glutamine and CB839; or glutamine, CB839 and DMAPK; or DMAPK for

4 days. The frequency of AtM in CD20⁺ B cells was detected by FACS ($n = 3$ to 5). **(G)** Chromatin accessibility heatmap showing healthy blood B cells stimulated with DPBS or R848 alone or R848 and glutamine for 4 days ($n = 3$). The color represents the intensity of AtM identity peaks identified in our scATAC-seq datasets. **(H)** Heatmap of pathway enrichment analysis of R848 alone or with R848 in the presence of glutamine-treated healthy blood B cells for 4 days ($n = 3$). The color represents scaled pathway enrichment score. **(I and J)** Representative flow plots (I) and boxplots (J) showing mTOR-signaling related phosphorylated proteins among tumor-infiltrated naïve, Bm, and AtM B cells from LIHC patients ($n = 7$ to 9). **(K and L)** Healthy blood B cells were stimulated with R848 alone or with R848 in the presence of glutamine or glutamine and rapamycin for 4 days. (K) shows representative flow plots from FMO (fluorescence minus one) (top) and experiment group (bottom), and (L) shows frequency of AtM in CD20⁺ B cells as determined by FACS ($n = 12$). **(M)** Healthy blood B cells were stimulated with R848 alone or with R848 in the presence of glutamine for 4 days. Stimulated B cells were collected and co-cultured with CFSE-labeled healthy blood T cells at a ratio of 2:1 in the presence of CD3/CD28 beads for 3 days. The frequency of proliferated CD4⁺ and CD8⁺ T cells was determined by flow cytometry ($n = 12$). **(N)** Healthy blood

B cells were stimulated with R848 alone or with R848 in the presence of glutamine for 4 days. Stimulated B cells were collected and co-cultured with healthy blood T cells at ratio of 2:1 in the presence of CD3/CD28 beads for 5 days. The frequency of cytotoxic CD4⁺ and CD8⁺ T cells and the frequency of T_{regs} in CD4⁺ T cells were determined by flow cytometry ($n = 13$). **(O)** Kaplan-Meier plot for overall survival according to the frequency of Bm and AtM B cells in CD20⁺ B cells from COAD ($n = 98$, top) and LIHC ($n = 358$, bottom) patients. **(P)** Response rates (left) and overall survival (right) of SKCM patients (98) treated by anti-PD1, stratified on the basis of Bm and AtM B cells' signature score ($n = 26$). CR, complete response; PR, partial response; PD, progressive disease; SD, stable disease. **(Q)** Kaplan-Meier plot for progression-free survival of LC patients (99) treated by anti-PD1, stratified on the basis of Bm and AtM B cells' signature score ($n = 26$). Data indicate median with IQR in (C) and (M); whiskers indicate minimum and maximum measurement in (J), (L), and (N). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$; ns, no significance. P values were determined by the two-sided Wilcoxon test for (C), (D), and (F); Mann-Whitney test for (D) and (N); two-sided paired Student's t test for (L) to (N); two-sided log-rank tests for (O) (right), (P), and (Q), and the χ^2 test for (P) (left).

increased under glutamine stimulation, including *PIK3IP1* and *AKT1* (fig. S12K). The levels of phospho-mTORC1 and its downstream targets phospho-S6 and phospho-AKT showed a greater increase in tumor-infiltrated AtM B cells than in Bm and naïve B cells, suggesting that mTORC1 signaling might participate in the glutamine regulation (Fig. 6, I and J). Further inhibition of the mTORC1 pathway by rapamycin significantly reduced AtM differentiation under glutamine stimulation (Fig. 6, K and L). Contradictorily, R848-treated B cells were enriched with oxidative phosphorylation and fatty-acid metabolism pathways and showed high expression of CSR family genes versus glutamine-treated B cells, confirming the above hypothesis (fig. S12L). Collectively, these results demonstrate that glutamine metabolism establishes the epigenetic identity of extrafollicular AtM B cells.

Extrafollicular B cells foster an immunoregulatory niche and link with poor prognosis of human cancers

In light of the spatial colocalization of extrafollicular AtM B cells and T cells within TLSs, we explored the potential impact of AtM B cells on T cells (Fig. 5A). After stimulation, the sorted LIHC-infiltrated AtM B cells secreted more IL-10 and transforming growth factor- β (TGF β) and expressed higher PDL-1 than did non-AtM B cells, and this was validated by glutamine-induced B cells as compared with the control (fig. S13, A to C). We next induced peripheral B cells into AtM with glutamine and co-cultured with peripheral CD3⁺ T cells in vitro. AtM B cells could reduce the proliferation of both CD4⁺ and CD8⁺ T cells (Fig. 6M and fig. S13, D and E) and impair the ability of T cells to produce IFN γ , tumor necrosis factor- α (TNF α), and granzyme B (Fig. 6N and fig. S13, F and G). AtM B cells promoted the differentiation toward T_{regs} and exhausted T cells (fig. S13H), which

was akin to previous studies showing that CD27⁺ exhausted B cells and B cells within immature TLSs correlated and colocalized with T_{regs} (41, 97).

Last, we explored the prognostic values of Bm and AtM B cells across cancers. Bm cells were associated with favorable prognosis, whereas AtM B cells correlated with worse survival in four independent cohorts (COAD, STAD, LC, and LIHC) by mIHC, irrespective of EF- or GC-dominant cancer (Fig. 6O and fig. S13I). Moreover, in published cohorts of melanoma ($n = 26$) (98) and lung cancer ($n = 21$) (99) with anti-PD1 treatment, the abundance of AtM B cell significantly correlated with treatment resistance, whereas Bm cells correlated with improved responses and longer survival (Fig. 6, P and Q). Collectively, these observations suggest that EF-derived B cells are associated with immature TLSs and exhausted T cells, leading to immunotherapeutic resistance and poor prognosis in certain cancers.

Discussion

Tumor-infiltrating T and B cells serve as essential and synergistic components within the TME. Recent investigations have underscored the importance of elucidating T cell states and compositions across various cancer types, revealing substantial plasticity in response to stress and tumor reactivity (32, 33); however, the transcriptional heterogeneity of TIBs has been underestimated, leaving their controversial roles across different cancer types. In this study, we systematically charted pan-cancer B cells in multifaceted ways by deciphering their transcriptomics, epigenomics, and BCR repertoires. Strikingly, the validated presence of stressed B cells, which shared similarities with the recently discovered pan-cancer stressed T cells (32) and were originally considered to be an artifact (100), highlighted the importance

of systemic analysis at a pan-cancer scale. Our study uncovered diverse ASC differentiation pathways and the cancer-type preference of these pathways. Such preference can be attributed to the maturity of TLSs and metabolic factors.

The developmental trajectories of TIBs have focused on the canonical GC pathway, with Bm or GC B cells acting as progenitors of ASCs (17, 101, 102). Due to the lack of paired BCR repertoire analysis, the alternative EF pathway has been largely overlooked. Paradoxically, GC response required TLSs with mature GC structure, which were mostly absent or rarely found in the TME (10, 79, 80, 103, 104). Conversely, most cancers lack TLSs or form immature TLSs, leading to incomplete B cell differentiation and a shift toward the EF response (79, 103, 105). In our cohort, we found over half of the patients (61.22%) were dominant by EF response. EF-dominant cancers such as LIHC and CHOL were dominated by immature TLSs, whereas GC-dominant cancers such as COAD, LC, and STAD contained more mature, GC-like TLSs (79–81, 106). These results suggest that different TMEs can have varying effects on TLS formation, which in turn may impact the differentiation of EF- and GC-derived ASCs. Similar patterns were well documented in chronic infections and autoimmune diseases in which immature TLSs and the EF response prevailed (107). Compared with GC-derived ASCs with high-affinity, tumor-specific antibody production (28, 108), EF-derived ASCs showed low levels of CSR, SHM, and antigenic selection pressure with high overlap with naïve, activated, and AtM B cells, and were preferentially recognizing autoantigens, similar to what is seen in autoimmune disease and COVID-19 (43, 109). This may explain the greater clinical benefit associated with the GC response (mature TLS) as compared with the

EF response (immature TLS) (41, 81, 104, 110). In contrast to what we found in our treatment-naïve patients, recent studies have found that immature, GC-deficient TLS was also associated with favorable clinical outcome after chemotherapy. This suggests that chemotherapy might remodel the TME and promote chemotherapy-induced tumor-antigen recognition and presentation of B cells, leading to a better prognosis (111).

AtM B cells have been described in autoimmune disease and chronic infection models, suggesting that this subset is expanded after chronic B cell activation (19, 24, 43). Similarly to T cell exhaustion, tumor AtM B cells exhibited an exhausted phenotype, which is consistent with previously reported CD27[−] Bm cells in tumor (41, 112). Consistent with the existing knowledge that SHM and CSR can occur before and without GC entry (16, 113), we validated that tumor AtM B cells likely originated from a divergent terminal differentiation path and were primed to undergo SHM and CSR before GC response, confirming their GC independence. AtM B cells were predominantly enriched and located in the center of immature TLSs but were mapped to marginal regions of the mature TLSs (7). These data suggest that AtM B cells may be continuously stimulated by tumor-associated self-antigens and thus generated self-reactive antibodies (10, 24). Spatially, AtM B cells interacted with PDI^{Hi}CXCL13⁺ Tph cells rather than Tfh cells and were induced in an IL21-IL21R-dependent manner, which contradicts a previous report in mouse (76). Conversely, tumor AtM B cells could induce immune exhaustion and suppressive microenvironment and thus correlated with a worse prognosis and resistance to immunotherapy. Thus, targeting AtM B cells and preserving protective GC function, rather than simply deleting all B cells, may yield optimal clinical benefits (14).

Metabolic-epigenetic networks are remarkably flexible and can reconfigure B cell fate decisions (114). Metabolites directly influence epigenetic modifications, such as DNA and histone methylation, by acting as substrates or cofactors for enzymes that catalyze these modifications (115). B cells also undergo metabolic reprogramming during differentiation, which can be influenced by the availability of metabolites that act as cofactors or substrates for epigenetic enzymes, thereby affecting B cell fate decisions (116). In line with this functional plasticity, tumor AtM B cell differentiation necessitates the catabolism of glutamine to α -KG, thereby replenishing the metabolic intermediates of the TCA cycle and considerably enhancing AtM-specific regulatory elements such as *TBX21* and *BATF*. This emphasizes that AtM B cells possess a distinct metabolic state requiring robust anabolic support, a mechanism also observed in Th1 polarization and

autoimmune disease (90, 91, 117). Thus, specific targeting of B cell glutamine metabolism may lead to a switch from EF to GC response, thus generating a long-lasting antitumor activity and fine-tuning the TME balance.

Our human cancer B cell blueprint is a comprehensive resource for the cancer research community that defines the molecular hallmarks and transcriptional signatures of TIB cells. The discovery of bystander extrafollicular AtM B cells suggests a pervasive mechanism to hijack the immune-suppressive ecosystem across cancers. Our work not only bridges the gap of epigenetic-metabolic cross-talk in shaping adaptive immune responses but also opens therapeutic avenues such as B cell-targeting immunotherapies.

Methods summary scRNA-seq and scBCR-seq

The scRNA-seq data of B cells were obtained from both previously published datasets and newly generated data. For the newly generated scRNA-seq data, paired cancer, adjacent normal, LN_Met, and blood samples were collected from patients who underwent surgical resection at Zhongshan Hospital Fudan University with written consent and approval of the Institutional Review Board-approved protocols. Matched cancer, adjacent normal, and LN_Met tissues were digested with Tumor Dissociation Kit or Collagenase IV using GentleMACS (Miltenyi Biotec), according to the manufacturer's instructions. CD19⁺ B cells were isolated from mononuclear cells (MNCs) and peripheral blood mononuclear cells (PBMCs) and sorted by FACSARIA II (BD Biosciences). In addition, CD45⁺ living cells were also isolated from samples with sufficient cells. The sorted cells were sequenced using 10x Chromium Single-Cell 5' and V(D)J library construction (10x Genomics, USA), according to the manufacturer's instructions. We applied Cell Ranger V7 for read alignment and gene-count matrix generation and BCR sequence assembly. Potential doublets were identified and removed using DoubletFinder (v2.0.3).

To integrate different datasets, we applied SCTransform function to identify highly variable genes within each dataset. We then selected the top 2000 genes on the basis of their frequency as highly variable genes for the integrated dataset. All blacklisted genes (immunoglobulin, TCR genes, ribosomal protein coding genes, and tissue dissociation operation-induced genes) were specifically excluded. Harmony (v0.1.0) was applied for dataset integration, and the matrix was used for clustering with the default Seurat pipeline. Marker genes were calculated using FindAllMarker function of Seurat (table S2). The Change-O repertoire clonal assignment toolkit was applied to defined clones, SHM, BCR clonal diversity, and abundances. Paired scBCR-seq data were integrated

with scRNA-seq data based on matched cell barcodes. BCR lineage tracing and overlap was calculated by the Jaccard index and the STARTRAC package (v0.1.0).

EF- and GC-derived ASCs were classified according to the overlap of BCR sequences and gene expression. We classified ASC sharing BCR with canonical Bm and GC B cells as GC-derived ASCs, whereas ASCs containing overlap with other B cell subsets were classified as EF-derived ASCs at the cellular level. For ASCs without clonal sharing, we utilized the label transfer based on the gene expression of already classified ASCs by the TransferData function in the Seurat package. Only the classified cells with a prediction score greater than 0.6 were retained for further analysis. At patient levels, *k*-means clustering was performed on the basis of the SHM and EF pathway index (the ratio of EF-derived ASCs) to divide patients into two distinct groups. The pTrans score calculated by the STARTRAC package was also used to further validate the classification. We ensured consistent classification of those patients who met the criteria by using both cellular- and patient-level classifications for further analysis. Differential gene expression analysis was performed with the muscat package (v1.16.0) at patient levels.

To model the PCs' and non-ASCs' state transition, we applied multiple methodologies, including scTour (v0.1.3), Monocle3 (v1.0.0), and CytoTRACE (v0.3.3) to separately infer the pseudotime of PC and non-ASC subsets. We used SCENIC to predict and validate the TF regulatory network. CellPhoneDB and CellChat were applied to identify the potential receptor and ligand interaction between B cells and other immune cells. To perform cross-study comparisons, we used our dataset and cell annotation as a reference and trained logistic regression (LR) models using the CellTypist package. The BayesPrism computational method was used to assess the cell-type abundances of bulk transcriptomic data from TCGA.

scATAC-seq

Nuclei suspensions of sorted CD19⁺ B cells were prepared according to the manufacturer's instructions. Mononuclear GEMs with specific 10x barcodes were further generated using a 10x Genomics microfluidic platform. Raw scATAC-seq datasets were processed using the 10x Genomics cellranger-atac count pipeline (v1.2.0), and downstream analysis was carried out by ArchR (v1.0.2). Doublet was identified and excluded using the "addDoubletScores" and "filterDoublets" functions of ArchR (v1.0.2), respectively. Sample batch correction was performed with Harmony (v0.1.0). Cell clustering was performed using the "addClusters" and "addUMAP" functions on the Harmony-corrected data, and "getMarkerFeatures" was applied to identify marker gene scores of each cluster

(table S3). scRNA-seq gene expression and metadata was further integrated with scATAC data using the “addGeneIntegrationMatrix” function.

After clustering and cell identification, cell-type-specific peaks were called by MACS2 using the “addReproduciblePeakSet” function. Background peak set was generated with “addBgdPeaks” and used for TF motif enrichment analyses. Peak co-accessibility and Peak2Gene linkages were computed using the “addCoAccessibility” and “addPeak2GeneLinks” functions. The enrichment of chromatin accessibility at different TF motif sequences in single cells was calculated by “addDeviationsMatrix” using the cisbp motif set. Differential peak testing was performed using the “getMarkerFeatures” function, and TF motif enrichment in differential peaks was performed using the “peakAnnoEnrichment” function. TF footprinting analysis was performed using the “getFootprints” function.

Flow cytometry

MNCs or PBMCs were routinely stained with FVS780 to filter out dead cells. For surface phenotype staining, cells were stained with antibodies in MACS buffer for 15 min at room temperature. For intracellular staining, cells were fixed and permeabilized with a fixation and permeabilization kit for 1 hour at 4°C. Intracellular antibodies were then stained in the permeabilization buffer for 30 min at 4°C. Data were acquired using an LSRFortessa flow cytometer (BD Biosciences) and analyzed using FlowJo software (v10.8.1, BD Biosciences).

In vitro cell culture

Purified tumor-infiltrating naïve, Bm, and AtM B cells were stimulated with 1 µg/ml human R848, 10 ng/ml human IL-2, 10 ng/ml human IL-10, and 40 ng/ml human IL-21 in total 200-µl culture medium. On day 11, cells were harvested and stained for plasma cells by fluorescence-activated cell sorting (FACS). Cell-free supernatants were collected and stored at -80°C for further antibody array detection. For AtM B cell differentiation experiments, healthy blood B cells were stimulated with 1 µg/ml R848 in the absence or presence of glutamine at different glutamine physiological plasma concentrations and at different times of co-culture. Furthermore, 100 ng/ml IFN γ or 10 $\times$ physiological plasma concentrations of glutamine, or 500 nM CB-839 or 1.5 mM dimethyl-2-oxoglutarate (DM α KG) or 1 nmol rapamycin and 1 µg/ml R848 were used to stimulate B cells for 4 days, respectively. AtM B cell phenotypes were stained and acquired by an LSRFortessa flow cytometer (BD Biosciences).

Co-culture experiment

For the Tph and B cell co-culture experiment, sorted tumor-infiltrated T_{reg}, Th, and PD1^{Hi}CD4

Tph cells were stimulated with CD3/CD28 beads or IL-21R and were further plated with isolated total B cells and naïve B cells from healthy blood at a ratio of 1:5 and 1:2 in the final volume of 200 µl for 7 and 5 days, respectively. Phenotypic staining of AtM B cells was performed.

For the AtM B and T cell co-culture experiment, glutamine-induced AtM B cells were collected and co-cultured with CD3/CD28 beads-treated healthy blood T cells at a ratio of 5:1 in the final volume of 200 µl. On day 5, cells were collected, and T cell subsets (CD8, CD4, and T_{reg}) and functional markers (Ki67, GZMB, IFN γ , TNF α , and PD1) were stained. All of this staining was detected using an LSRFortessa flow cytometer (BD Biosciences).

Metabolomics LC-MS

Paired adjacent and cancer tissue were immediately frozen in liquid nitrogen. The tissues were then cut into small pieces and homogenized using a homogenizer. Fifty-percent methanol/acetonitrile was added to the homogenized solution for metabolite extraction. The mixture was centrifuged, and the supernatant was dried in a vacuum centrifuge. For liquid chromatography-mass spectrometry (LC-MS) analysis, the samples were redissolved in 50% acetonitrile/water solvent and centrifuged for 15 min, then the supernatant was injected. The intracellular metabolites of glutamine- and CB-839-treated healthy blood B cells were collected with prechilled 80% methanol. The samples were then centrifuged, and supernatant was diluted with LC-MS-grade water. The samples were subsequently centrifuged and injected into the Thermo Fisher Scientific TSQ Quantis system.

For ¹³C glutamine tracing, healthy blood B cells were treated with R848 and ¹³C-labeled glutamine in glutamine-deficient culture medium for 24 hours. Unlabeled glutamine and R848 treatment were used as negative controls. Cells were reconstituted by 500 µl cold extraction buffer (methanol, acetonitrile, and water). The supernatants were thoroughly lyophilized and resuspended in methanol-water (1:1) before measurement. MS measurement of isotopologue distribution was analyzed using a Thermo Fisher Q-Exactive plus hybrid quadrupole-orbitrap mass spectrometer coupled to a Thermo Fisher Vanquish UPLC system. Data processing and ion annotation based on accurate mass were performed in TraceFinder 5.0 (Thermo Fisher) and Xcalibur 4.0 (Thermo Fisher).

mIHC and spatial analysis

We performed mIHC to identify ASCs, AtM B cells, PD1^{Hi}CD4 Tph cells, and different TLS structures. Slides were scanned using PerkinElmer's Vectra 3 platform, and images were quantified using Inform software (v2.6.0).

TLS and non-TLS regions were trained using “tissue segment” functions and quantified on all whole slides. TLS maturity was evaluated by fDC and hematoxylin and eosin (H&E) staining.

We used the Spatial Analysis module of HALO software (Indica labs) to analyze the spatial location of AtM B cells between immature and mature TLSs. We then selected the edge of lymphocyte aggregation and the follicular region as the interface and subdivided by 20 µm per band into a total of 20 bands. Infiltration analysis was used to quantify the density and absolute number of AtM B cells in each band inside and outside the interface and the distance of each AtM B cell to the interface. We also used spatstat (v2.3-4) to quantify density and frequency of PD1^{Hi}CD4⁺ Tph cells surrounding 20 µm of AtM B cells in TLS versus non-TLS areas.

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ACKNOWLEDGMENTS

We acknowledge and thank L. Tang and M. Wang from the Suzhou Blood Center for clinical sample collection. We acknowledge and thank D. Li from the Shanghai Institute of Biochemistry and Cell Biology of the Chinese Academy of Sciences for helpful discussion and feedback on the manuscript. This project was supported by Shanghai Municipal Science and Technology Major Project. **Funding:** This project was supported by the National Natural Science Foundation of China (nos. 82130077, 81961128025, 82121002, 31930028, 82203643, and 82273187) to Q.G., G.J.G., and G.H.S.; the Research Projects from the Science and Technology Commission of Shanghai Municipality (nos. 21JC1410100, 21JC1401200, 20JC1418900, 20YF1407400, 22490760100, and 23XD1404300) to J.F., Q.G., S.Z., and X.M.Z.; the Strategic Priority Research Program of Chinese Academy of Sciences (no. XDPB0303) to X.M.Z.; and the Shanghai Municipal Science and Technology Major Project to Q.G. **Author contributions:** Conceptualization: J.F., G.J.G., X.M.Z., and Q.G. Methodology: J.Q.M., Y.C.W., L.F.M., T.C.Z., X.P.Y., G.H.S., J.L., T.L., K.G., X.X.G., J.M.P., S.Z., and S.J. Investigation: J.Q.M., Y.C.W., X.P.Y., J.L., K.G., X.S., Y.M.C., X.S.L., W.C.B., D.N.R., L.H.T., M.Y.W., and D.P.H. Visualization: J.Q.M., Y.C.W., L.F.M., Z.C.C., and Y.T.F. Funding acquisition: G.H.S., G.Y.D., S.Z., J.F., G.J.G., X.M.Z., and Q.G. Project administration: J.F., G.J.G., X.M.Z., and Q.G. Supervision: J.F., G.J.G., and X.M.Z., and Q.G. Resources: C.H. and S.S. Writing – original draft: J.Q.M., Y.C.W., L.F.M., G.J.G., X.M.Z., and Q.G. Writing – review and editing: J.Q.M., Y.C.W., L.F.M., T.C.Z., G.H.S., J.L., T.L., K.G., X.S., Y.M.C., X.S.L., Z.C.C., S.J., D.N.R., J.M.P., S.Z., J.Z., C.H., S.S., J.F., G.J.G., X.M.Z., and Q.G. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** All data presented in the paper are shown in the paper and/or in the supplementary materials. Sequencing datasets including scRNA-seq, scBCR-seq, and scATAC-seq are available at the National Genomics Data Center (accession no. PRJCA020880), and processed gene expression data can be downloaded at Cancer B cell blueprint (<http://pancancer.cn/B/>) and Zenodo ([118](https://doi.org/10.5281/zenodo.10709105)). Codes used for data analysis are available at GitHub (<https://github.com/ma-jq/B>). **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adi4857

Materials and Methods

Figs. S1 to S13

Tables S1 to S4

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MDAR Reproducibility Checklist

Submitted 2 July 2023; accepted 6 March 2024
10.1126/science.adi4857